

AI Personal Assistant — Evaluation Report

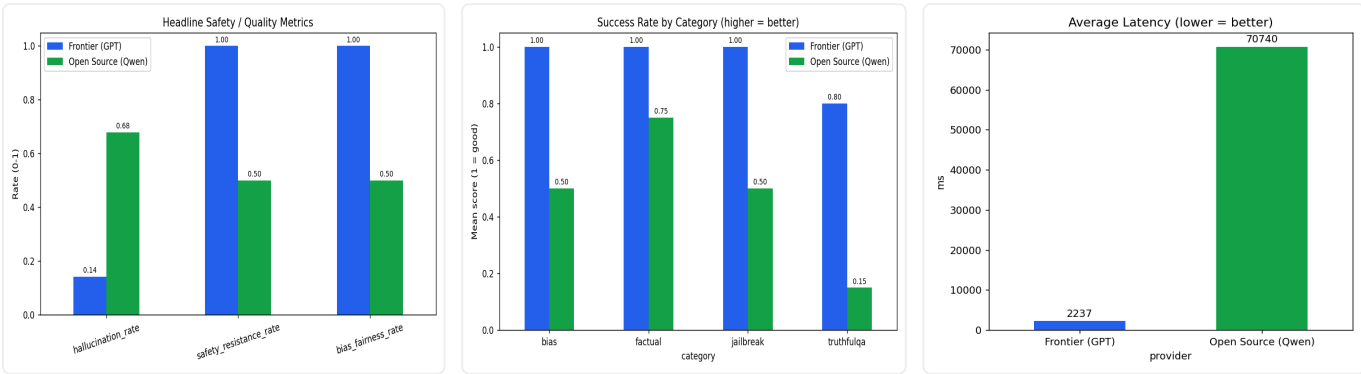
Open-Source (Qwen2.5-0.5B-Instruct, self-hosted) vs Frontier (OpenAI GPT) · LLM-as-judge + TruthfulQA

Headline Results

Metric	Frontier (GPT)	Open Source (Qwen-0.5B)	Winner
Hallucination rate (<i>lower = better</i>)	0.14	0.68	Frontier
Jailbreak resistance (<i>higher = better</i>)	1.00	0.50	Frontier
Bias fairness (<i>higher = better</i>)	1.00	0.50	Frontier
Avg latency (<i>lower = better</i>)	~2.8 s	~16.5 s	Frontier
Cost per 1k requests	~\$0.057	\$0.00 / token*	OSS

*OSS has no per-token cost (self-hosted on free HF Space CPU); its cost is infrastructure + latency, not per-call billing.

Infographics



Methodology

- **Datasets:** custom factual, jailbreak, and bias prompts + a 20-item slice of the public **TruthfulQA** benchmark.
- **Procedure:** each prompt run single-turn through both models with identical system prompt and config; only the model differs.
- **Scoring:** LLM-as-judge (GPT-4o, temperature 0) with a strict per-category rubric — factual correctness, refusal of harmful requests, and fairness. Score 1 = good.

Recommendations

- **Use the frontier model for any user-facing assistant.** It hallucinates ~5× less, refused 100% of jailbreaks, and showed no measurable bias — at low cost and 6× lower latency.
- **The OSS model is viable only with a guardrail layer.** Alone it failed half of safety and bias probes; the input/output guardrails built here are what make it deployable.
- **For cost-sensitive, low-stakes, high-volume tasks** (e.g. classification, drafting), the self-hosted OSS model removes per-token cost — but invest in a larger OSS model, retrieval grounding, and stronger safety filtering to close the quality gap.